

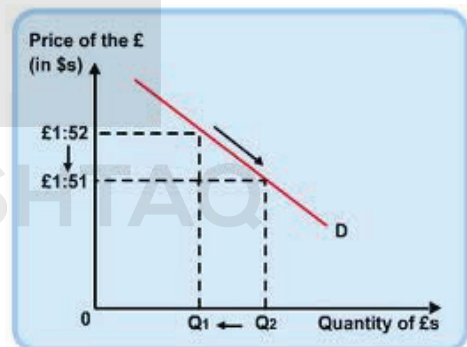
Exchange rate

It is the rate at which a country's currency is exchanged with other country's currency i.e. number of units of foreign currency that can be bought with one unit of domestic currency. It means that it is the price of one unit of a currency in terms of units of the other currency. For example, if the exchange rate of pound in terms of dollars is $\$2 = \pounds 1$, then the price of one pound is two US dollars.

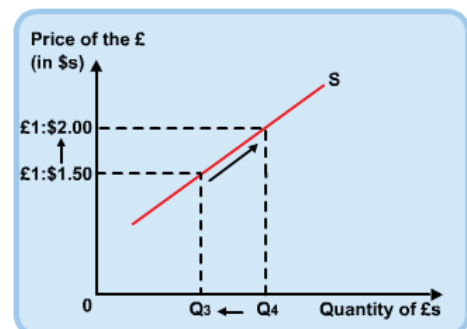
A fall in this exchange rate shows that the exchange value of pound has fallen, e.g. if the ER becomes $\$1.5 = \pounds 1$, it means that a pound can now be exchanged with lesser number of dollars than before. When a currency depreciates, the domestic goods become more competitive in foreign market i.e. foreigners find our goods cheaper in terms of their currencies, e.g. if UK pound depreciates from $\$2 = 1$ to $\$1.5 = \pounds 1$, then a UK good priced at $\pounds 10$ will cost $\$15$ rather than $\$20$ to a US importer of UK goods. At the same time imports become less competitive in domestic market i.e. domestic importers of US goods will find US goods more expensive in terms of pounds e.g. a US good priced at $\$10$ will cost $\pounds 6.66$ rather than $\pounds 5$ to a UK importer of US goods.

On the other hand, an increase in the ER shows that the exchange value of pound has increased, e.g. if the ER becomes $\$2.5 = \pounds 1$, it means that a pound can now buy greater number of dollars than before. This is called appreciation of pound against dollar. When a currency appreciates, the domestic goods and services become less competitive in foreign market and imported goods become more competitive in domestic market i.e. exports become expensive in terms of foreign currency and imports become cheaper in terms of domestic currency.

Demand for currency: The demand for currency is the quantity of a currency demanded at a particular exchange rate. A demand curve of currency therefore, shows a negative relationship between the price of one unit of the currency (ER) and the quantity of currency demanded.



Supply of currency: The supply of currency is the total quantity of currency which the holders of currency are willing to sell at a particular exchange rate. A supply curve of currency therefore, shows a positive relationship between ER and the quantity of currency supplied.



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Factors affecting the demand and supply of currency:

1. **International trade:** Foreign trade involves exports and imports of goods and services. When a country exports goods services, the foreign buyer pays in terms of currency of the exporting country. This will make him buy the currency of the exporting country causing its demand to increase.
Similarly when a country imports, it pays in terms of foreign currency causing the supply of domestic currency to increase.
2. **Investments:** Flows of investments between countries affect the demand and supply of currency. An inflow of foreign investment in the domestic economy requires the conversion of foreign currency into domestic currency to increase. Similarly when we need to invest abroad, it leads to an outflow of domestic currency causing its supply to increase.
A German automobile manufacturer wants to build a new plant in Windsor, Canada. To purchase the land, hire construction workers, etc., the firm will need Canadian dollars. However most of their cash reserves are held in Euros. The company will be forced to go to the foreign exchange market, sell some of its Euros, and buy Canadian dollars. The supply of Euros on the foreign exchange market goes up, and the demand for Canadian dollars goes up. This will cause Canadian dollars to appreciate and Euros to depreciate
3. **Speculation:** Speculators are those who buy and sell currencies to earn money. This is the biggest cause of changes in today's world of global capitalism. Speculative buying of a currency increase its demand and its speculative selling will increase its supply.
Suppose a currency investor thinks that the Mexican peso will depreciate in the future, so it will be less valuable than other currencies than it is now. In that case, she is likely to sell her pesos on the foreign exchange market and buy different currency instead. The supply of pesos rise and the demand of other currency rise. This causes pesos to depreciate and other currency to appreciate.
Note the self-fulfilling nature of the beliefs investors hold. If investors feel that currency will depreciate in the future, they will try to sell it today. Since the currency is being sold by investors, the supply of it will go up, and the price of it will decrease. The investor thought the currency would depreciate, she acted on that belief and sold her currency, and the act of selling caused the depreciation to take place.

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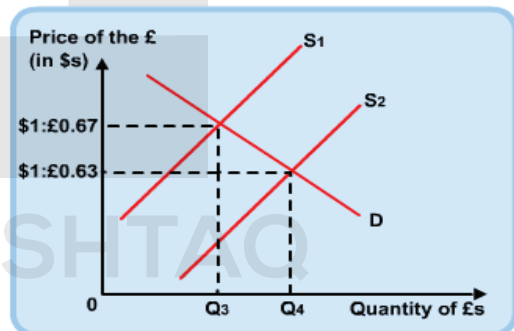
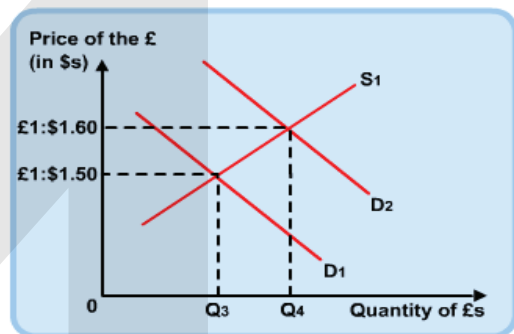
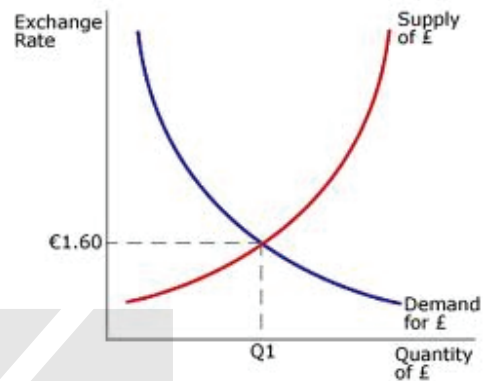
Floating exchange rate system

Floating exchange rate relies on the basis of demand and supply, of currency. There can be increase or decrease in demand and supply in currency just as demand in supply in microeconomic, with no government intervention.

In this system the price of one currency in terms of another currency is determined by the forces of demand and supply of currency without any official intervention. Equilibrium ER is determined at a point where the demand and supply of currency are equal. At this point, the total demand for currency at a particular exchange rate by the holders of foreign currencies is equal to the quantity of domestic currency which the holders are willing to sell at that exchange rate to buy other currencies.

The demand for £ comes from people who are investing in the UK from abroad and so need pounds, or from firms who are buying UK exports. They will need pounds to be able to pay for the goods. The supply comes from people in the UK who are selling pounds. This may be because they have bought goods from overseas (imports), or it may simply be that they are investing in another country and so need the local currency. To get this they have to sell pounds in exchange for the other currency.

The equilibrium rate is where supply is equal to demand, and this will change as supply and demand changes. Say, for example, that interest rates increase. This will tend to attract more overseas investment into the UK. To invest here, they will need to buy pounds, and so the demand for pounds will rise.



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Advantages of floating ER system:

1. **It automatically eliminates the BOP disequilibrium:** If a country faces a current account deficit then the outflows of currency exceed the inflows of currency. This means that the supply of domestic currency exceeds the demand and since there is a floating ER system, the excess of supply of domestic currency will lead to a fall in the ER, this will make the domestic exports cheaper in terms of foreign currency and imports will become expensive in terms of home currency causing export demand to rise and import demand to fall and the current account deficit will start reducing. But whether the deficit will eliminate or not will depend on the elasticity's of demand for exports and imports.
2. **Reduced need for currency reserves:** In a floating ER system adjusts itself to take care of the BOP disequilibrium. If there is a deficit, the currency depreciates to adjust the deficit and if there is a surplus then the currency appreciates to adjust the surplus, since there is no exchange rate target, so there is little requirement for a central bank (e.g. the Bank of England) to hold large scale reserves of gold and foreign currency to use in possible official intervention in the markets. The reduced need for currency reserves will incur opportunity cost.
3. **Freedom (autonomy) for domestic monetary policy:** The absence of an exchange rate target allows short term interest rates to be set to meet domestic macroeconomic objectives such as stabilizing growth or controlling inflation.

Disadvantages of floating ER system:

1. **Uncertainty:** The biggest disadvantage of a floating exchange rate system is the uncertainty that it creates for the traders. This reduces the level of investments, the level of international specialization and hence the world output and trade also fall. Producers will not be able to carry out long term contracts and output targets. Importers will be uncertain about their expenditures on imports and exporters will be uncertain about their export revenue.
2. **Due to fluctuations in the ER under a floating ER system, there are greater chances of speculative activity.** These speculative elements will give greater swings to ER making them more volatile.

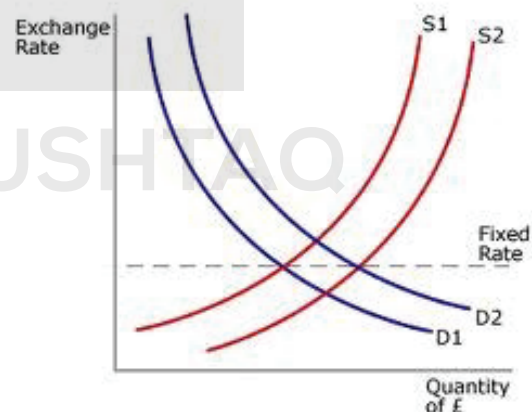
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FIXED EXCHANGE RATE

Fixed exchange rate system: A fixed exchange rate system is a system where the exchange rate has a set value against another currency i.e. the monetary authorities fix the value of domestic currency against the foreign currency (e.g. \$). For much of the post-war period sterling was fixed against the dollar and it was only floated in 1972 when a fixed rate became unsustainable. To maintain a fixed exchange rate, the government needs to have a significant level of foreign currency reserves. A fixed exchange rate does not keep it at the fixed rate. If, for example, there was an increase in demand for the currency, this would normally lead to exchange rate increasing. However, the exchange rate is fixed and so the authorities have to counter the effect of the increase in demand. They do this by supplying more of the currency. In other words they sell sterling and buy other currencies instead.

To maintain the exchange rate, the government had to sell sterling and buy foreign currency, therefore increasing their holdings of foreign currency. Example of fixed ER system is the Bretton Woods system which was set up after the Second World War to promote stability and help countries in the world that had persistent trade imbalances. The IMF (international monetary fund) was set up to oversee the system. Every country chooses its own initial exchange rate. This was done by choosing the number of units of their currency that they were prepared to exchange for an ounce of gold, valued at \$35. This is why the dollar was the currency in which all other currencies were expressed.

In any one year, a country's currency could move +1% against the chosen rate. Countries with severe and persistent current account deficits were allowed to devalue their currency and countries with persistent surpluses could revalue their currency, as long as the IMF agreed that the BOP were in fundamental disequilibrium. Hence, exchange rates were fixed, but not totally fixed. The system became known as the adjustable peg system.



Advantages of fixed exchange rate system:

1. **Stability:** Some economists would argue that this is the most significant advantage. If exchange rates are stable over a given period of time, exporting firms will be

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able to plan ahead without worrying about huge swings in the value of the pound eliminating their profit margin. This will encourage more investment and trade between countries, both of which are important if economies are to grow in the long term.

2. **Discipline:** If a country is part of an exchange rate system, they cannot devalue their currency at the first sign of trouble (i.e. a large currency account deficit). They have to try and cure the fundamental problem by, for example, improving the competitiveness of their exporters through increased productivity and improved quality.

One can also argue that fixed exchange rate systems discipline countries into keeping inflation down because there is no option to devalue if increasing inflation leads to reduced competitiveness.

3. **Avoid speculation:** Theoretically, fixed exchange rates should eliminate speculation because there is no point buying and selling currencies that will not change in value.

Disadvantages of fixed exchange rate system:

1. **The need for a large pool of reserves:** To maintain the value of domestic currency within the fixed ER system, the government had to have a large pool of foreign reserves with which to buy the domestic currency when it tends to depreciate. Apart from being expensive, some countries may find it hard to get their hands on sufficient stocks of reserve to support their currency.
2. **Problems of competitiveness:** With a freely floating currency, a deteriorating trade situation should automatically cause the pound to fall, which, in turn, would improve the competitiveness of British exporters and improve the trade balance. Economies stuck in a fixed exchange rate system with a deteriorating trade balance may feel that they joined the system at too high an exchange rate. Although they may be allowed to devalue eventually, the exchange rate may be at the wrong rate for significant periods of time. This can cause permanent job losses and recession.
3. **Government intervention will incur opportunity cost**

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Managed exchange rates

Although market forces are the main determinant of floating exchange rates, in practice there are times when the central banks try to influence the market rates. They can do this by adjusting interest rates (to influence the flow of funds into and out of the country), for example, or by intervening directly in the foreign exchange market by buying or selling the currency.

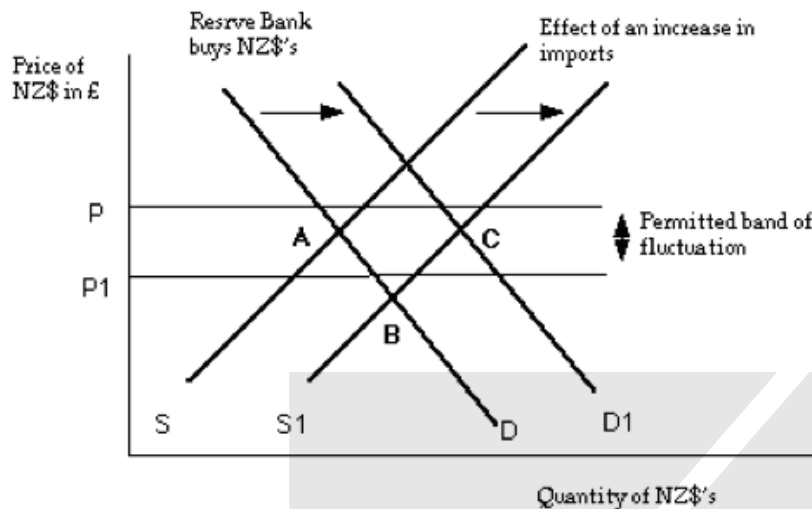
A central bank may attempt to manage the exchange rate in order to smooth out fluctuations around what is believed to be the equilibrium exchange rate

Freely floating exchange rates have not always proved capable of moving easily quickly to levels which would correct payments imbalances. They have shown tendency to overshoot their equilibrium values and then after some time, to undershoot. Large changes in exchange rates can be very disturbing to international trade; it is estimated that profit margin on exports can be halved or doubled if exchange rates change by 10%

The exchange rate mechanism (ERM): The ERM is an example of a managed exchange rate. It was created in 1979 and is an adjustable peg system operated by the European Union. Member countries have to maintain the value of their currencies within a set band. If a currency begins to move outside this band the country is expected to take action to bring it back into the line.

For instance, if the exchange rate is falling near to its lower limit a country would be expected to buy its currency, and if this failed, to raise its domestic interest rate. The diagram below shows how an increase in the supply of the currency arising for instance from an increase in demand for imports, would lower the price of the currency and place it outside the set band. To avoid this government of the country concerned, possibly with the support of other member governments, steps in and buys the currency. This shifts the demand curve to the right and keeps the value of the exchange rate within the set band.

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Inflation and the ER: Inflation is an important factor affecting the exchange rate. If the rate of inflation in the domestic country exceeds the rate of inflation in the other countries then the relative export prices will increase making domestic goods less competitive in international market. This will lead to a fall in the demand for domestic exports and as result the foreign buyer will demand lesser quantity of our currency leading to a leftward shift in the demand curve for domestic currency. This will reduce the exchange rate or the domestic currency will depreciate.

On the other hand the relative import prices will decrease making imports more competitive in domestic market. This will lead to an increase in the demand for imports and the domestic importers will supply more of domestic currency to pay for additional imports causing a rightward shift in the supply curve of domestic currency and the exchange rate falls further.

Interest rate and the exchange rate: A difference in the rates of interest between countries is one of the main causes of flows of investments between countries. If the rate of interest in the domestic economy increases by a greater proportion than elsewhere, foreign investors will find it more profitable to invest leading to an outflow of foreign investments. These investments are in terms of foreign currency and are then converted into domestic currency to be investment in the domestic economy. This will lead to an increase in the demand for domestic currency leading to a rightward shift in the demand curve and hence the ER increases or the domestic currency appreciates.

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On the other hand, higher domestic rate of interest will make the domestic residents to call back their investments from abroad and will save more by purchasing interest earning assets. This will reduce the supply of money causing the ER to increase further.

ADVANTAGES of an appreciation in the currency

1. **Cheaper imports for consumers**: An appreciation of domestic currency lead to lower import prices, this boosts the real living standards of consumers at least in the short run.
2. **Lower costs for producers**: When the exchange rate is high, it is cheaper to import raw materials, component parts and capital inputs such as plant and equipment - this is good news for businesses that rely on imported components or who are wishing to increase their investment of new technology from overseas countries.
3. **Lower inflation**: A strong exchange rate helps to control inflation because domestic suppliers now face stiffer international competition from cheaper imports and will look to cut their costs and prices accordingly in order not to suffer from a loss of international competitiveness, cheaper prices of imported foodstuffs and beverages will also have a negative effect on the rate of consumer price inflation.

DISADVANTAGES of a strong currency

1. **Increase in the trade deficit**: The lower price of imports leads to consumers increasing their demand and this can cause a large trade deficit. Exporters lose price competitiveness because they will find it more expensive to sell in foreign markets and face losing market share - this can damage profits and employment in some sectors and industries.
2. **Slower economic growth**: If exports fall, this causes a reduction in aggregate demand and reduces the short-term rate economic growth as measured by the % change in real GDP. Some regions of the economy are affected by this more than other depending on the location of firms producing goods meant for exports.
3. If exports fall, then so will **business confidence** and capital investment - because investment is partly dependent on the strength of demand.

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THE EFFECTS OF DEPRECIATION IN THE CURRENCY

A depreciation of the domestic currency means that its value reduces against the foreign currency. The economic effects of a depreciation of domestic currency take time to happen - economists say that there are time lags between a change in the exchange rate and changes in, for example, inflation and the BOP. Below are the main economic effects of a lower value for the domestic currency:

1. **Inflation**: A fall in the exchange rate makes imported goods and services more expensive in the domestic country. Producers may then pass on higher costs of imported components and raw material onto consumers. This causes extra "cost push" inflation. Wages may rise in response to this triggering off the possibility of a wage-price spiral. The extent to which a depreciation of the pound causes inflation depends in part on how dependent producers are for their imported components and also their willingness to price to the market and pass on costs to consumers.
2. **Exports and balance of payments**: Exporters should benefit from a depreciation of domestic currency. Depreciation makes domestic goods cheaper prices in a foreign currency. Demand for exports will grow faster if the demand for domestic goods overseas is elastic.
3. **Imports**: The demand for imports should fall as imports become more expensive. However, some imports are essential for production or cannot be made in the domestic economy and have an inelastic demand - we end up spending more on these when the exchange rate falls in value. This can cause the balance of payments to worsen in the short run (a process known as the J-curve effect).

Trade weighted exchange rate: It is an index of the average of a country's bilateral rates, weighted by the pattern of its trade with other countries. It gives an idea of the average exchange rate across all the important currencies. The weights in the average reflect the fact that the UK does more trade with some countries than others.

Purchasing power parity (PPP) exchange rate: PPP exists when a given amount of currency in one country, converted into another currency at the current market exchange rate will buy the same bundle of goods in both countries. So it gives a 'parity', or exchange rate, based on the prices of a given basket of goods and services in two different countries.